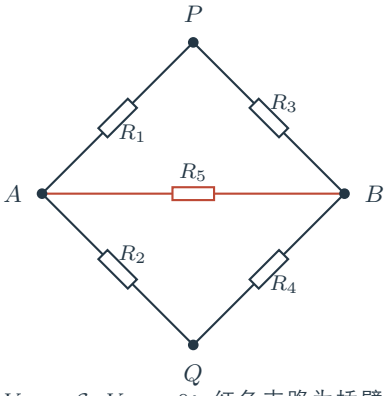
M4-032 一般电桥：保留桥臂的有符号电流



$V_P = \mathcal{E}$, $V_Q = 0$；红色支路为桥臂

$$I_5 = (V_A - V_B)/R_5$$

$$\frac{V_A - \mathcal{E}}{R_1} + \frac{V_A}{R_2} + I_5 = 0$$

$$\frac{V_B - \mathcal{E}}{R_3} + \frac{V_B}{R_4} - I_5 = 0$$